

Mastering Quantum Mechanics — Problem 9.5

A particular Hermitian operator

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Problem 9.5

A particular Hermitian operator. Consider $\hat{\Omega} = \hat{x}\hat{p} + \hat{p}\hat{x}$. (1) Is it Hermitian? (2) For a real normalizable $\psi(x)$ vanishing at $\pm\infty$, what is $\langle\hat{\Omega}\rangle$? (3) Write $\hat{\Omega}$ in terms of $\hat{a}, \hat{a}^\dagger$. (4) Evaluate $\langle\varphi_n|\hat{\Omega}|\varphi_n\rangle$. (5) Show any operator $[\hat{\Omega}', \hat{H}]$ has zero expectation in an oscillator eigenstate, and connect to (4) (and Exercise 9.12).

Hints & Orientation

- (1) Use $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$ with $\hat{x}, \hat{p}$ Hermitian; the symmetrized product is built precisely to be Hermitian (note $\hat{x}\hat{p}$ alone is *not*).
- (2) Put $\hat{p} = -i\hbar d/dx$ and integrate by parts; the boundary terms vanish and $\int x(\psi^2)' dx = -\int \psi^2 dx$.
- (3) Insert Eq. (9.3.12); the diagonal pieces cancel and only the $\hat{a}^2, \hat{a}^{\dagger 2}$ terms survive.
- (4) $\hat{a}^2, \hat{a}^{\dagger 2}$ change n by ± 2 , so they are orthogonal to φ_n .
- (5) $\langle\varphi_n|[\hat{\Omega}', \hat{H}]|\varphi_n\rangle = 0$ because $\hat{H}\varphi_n = E_n\varphi_n$. Then exhibit $\hat{\Omega}$ itself as such a commutator.

Solution

(1) Hermiticity

$\hat{\Omega}^\dagger = (\hat{x}\hat{p})^\dagger + (\hat{p}\hat{x})^\dagger = \hat{p}^\dagger\hat{x}^\dagger + \hat{x}^\dagger\hat{p}^\dagger = \hat{p}\hat{x} + \hat{x}\hat{p} = \hat{\Omega}$. **Yes, $\hat{\Omega}$ is Hermitian.** (Symmetrizing is what makes it so: $\hat{x}\hat{p}$ by itself satisfies $(\hat{x}\hat{p})^\dagger = \hat{p}\hat{x} \neq \hat{x}\hat{p}$.)

(2) Expectation in a real state

With $\hat{p} = -i\hbar d/dx$ and ψ real, $\hat{\Omega}\psi = -i\hbar(x\psi' + (x\psi)') = -i\hbar(2x\psi' + \psi)$, so

$$\langle\hat{\Omega}\rangle = -i\hbar \int \psi(2x\psi' + \psi) dx = -i\hbar \left(2 \int x\psi\psi' dx + \int \psi^2 dx \right). \quad (1)$$

Since $\int x\psi\psi' dx = \frac{1}{2} \int x(\psi^2)' dx = -\frac{1}{2} \int \psi^2 dx$ (boundary terms vanish), the bracket is $-\int \psi^2 + \int \psi^2 = 0$:

$$\boxed{\langle\hat{\Omega}\rangle = 0} \quad \text{for any real } \psi \text{ vanishing at } \pm\infty. \quad (2)$$

(3) In terms of ladder operators

From Eq. (9.3.12), $\hat{x} = \frac{L_0}{\sqrt{2}}(\hat{a} + \hat{a}^\dagger)$, $\hat{p} = \frac{i}{\sqrt{2}}\frac{\hbar}{L_0}(\hat{a}^\dagger - \hat{a})$. A short computation using $[\hat{a}, \hat{a}^\dagger] = 1$ gives $\hat{x}\hat{p} = \frac{i\hbar}{2}(1 - \hat{a}^2 + \hat{a}^{\dagger 2})$ and $\hat{p}\hat{x} = \frac{i\hbar}{2}(\hat{a}^{\dagger 2} - \hat{a}^2 - 1)$, hence

$$\boxed{\hat{\Omega} = \hat{x}\hat{p} + \hat{p}\hat{x} = i\hbar(\hat{a}^{\dagger 2} - \hat{a}^2)}. \quad (3)$$

(One checks $\hat{\Omega}^\dagger = -i\hbar(\hat{a}^2 - \hat{a}^{\dagger 2}) = \hat{\Omega}$, Hermitian as found.)

(4) Expectation in an eigenstate

$\hat{a}^{\dagger 2}\varphi_n \propto \varphi_{n+2}$ and $\hat{a}^2\varphi_n \propto \varphi_{n-2}$, both orthogonal to φ_n , so

$$\boxed{\langle \varphi_n | \hat{\Omega} | \varphi_n \rangle = i\hbar(\langle \varphi_n | \hat{a}^{\dagger 2} | \varphi_n \rangle - \langle \varphi_n | \hat{a}^2 | \varphi_n \rangle) = 0}. \quad (4)$$

(5) Commutator argument

For any $\hat{\Omega}'$, since $\hat{H}\varphi_n = E_n\varphi_n$,

$$\langle \varphi_n | [\hat{\Omega}', \hat{H}] | \varphi_n \rangle = \langle \varphi_n | \hat{\Omega}' \hat{H} | \varphi_n \rangle - \langle \varphi_n | \hat{H} \hat{\Omega}' | \varphi_n \rangle = E_n \langle \varphi_n | \hat{\Omega}' | \varphi_n \rangle - E_n \langle \varphi_n | \hat{\Omega}' | \varphi_n \rangle = 0. \quad (5)$$

Our operator is of exactly this form: using $[\hat{x}^2, \hat{H}] = \frac{1}{2m}[\hat{x}^2, \hat{p}^2] = \frac{i\hbar}{m}(\hat{x}\hat{p} + \hat{p}\hat{x})$,

$$\hat{\Omega} = \hat{x}\hat{p} + \hat{p}\hat{x} = \frac{m}{i\hbar}[\hat{x}^2, \hat{H}] = \left[\frac{m}{i\hbar}\hat{x}^2, \hat{H} \right]. \quad (6)$$

So $\langle \hat{\Omega} \rangle_{\varphi_n} = 0$ follows immediately — this is the same mechanism Exercise 9.12 exhibits (a commutator with $\hat{H}$ produces a Hermitian operator whose eigenstate expectation value must vanish), explaining part (4).

Checks

- **Consistency of (2) and (4)/(5).** Three independent routes (real-state integration, ladder structure, commutator-with- $\hat{H}$) all give 0.
- **Hermiticity.** $\hat{\Omega} = i\hbar(\hat{a}^{\dagger 2} - \hat{a}^2)$ is manifestly Hermitian, consistent with part (1).
- **Classical reading.** $\hat{\Omega} = \hat{x}\hat{p} + \hat{p}\hat{x}$ is $2m\overline{\dot{x}x} = m d\langle x^2 \rangle/dt$; in a stationary state nothing evolves, so its average vanishes.